TOMCAT WEBSERVER

What is Apache Tomcat

Apache Tomcat is a Java application server/web container. More specifically, it implements Java/Jakarta web technologies such as Servlet, JSP, WebSocket, etc.

TOMCAT DIRECTORIES

apache-tomcat-11.x.x/
├── bin/
├── conf/
├── lib/
├── logs/
├── temp/
├── webapps/
├── work/

1. **bin/ — Tomcat startup and control**

bin used for start, stop, and manage Tomcat.

1. startup.sh ----------> To start tomcat in the background
2. shutdown.sh -------> To stops tomcat
3. catalina.sh start ---> To start terminal in the background
4. catalina.sh run -----> Starts Tomcat in the foreground also see the logs on terminal

2. **conf/ — Tomcat configuration**

Contains the configuration files of Tomcat.

1. server.xml → Main Tomcat configuration
2. web.xml → Web application configuration
3. tomcat-users.xml → Users and roles
4. context.xml → Application context configuration

3. **lib/ — Libraries/Jar files**

Contains JAR files required by Tomcat.

Ex. Mysql-connector.jar

4. **logs/ — Log Files**

Contains Tomcat's log files.

Logs are useful for finding:

- Application errors
- Startup problems
- Shutdown information
- HTTP requests/access information

5. **temp/ — Temporary Files**

Tomcat uses this directory to store temporary files while it is running.

For example, Tomcat may create temporary files during:

- Application processing
- File handling
- JSP processing
- Other temporary operations

6. **webapps/ — Applications**

It contains the web applications deployed on Tomcat. (war)

7. **work/ — Working/Generated Files**

Tomcat uses this directory for working files generated while applications are running.

APPLICATION DEPLOYMENT ON TOMCAT

1.Installation of java

1. apt install openjdk-21-jdk -y
2. java –version

2.Installation of tomcat

1. wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.57/bin/apache-tomcat-10.1.57.tar.gz
2. tar -xzf apache-tomcat-10.1.57.tar.gz
3. cd apache-tomcat-10.1.57/
4. cd webapps
5. wget https://s3-us-west-2.amazonaws.com/studentapi-cit/student.war
6. cd ..
7. cd bin
8. ./startup.sh

3.To check the application

http://public_ip:8080/student

CHANGING PORT OF TOMCAT

1. cd apache-tomcat-10.1.57
2. cd conf
3. vim server.xml
   Search the line <Connector port="8080" protocol="HTTP/1.1"
   Change port to 9090
4. cd ../bin
5. ./catalina.sh stop
6. ./catalina.sh start

Check Application

1. http://public_ip:9090
